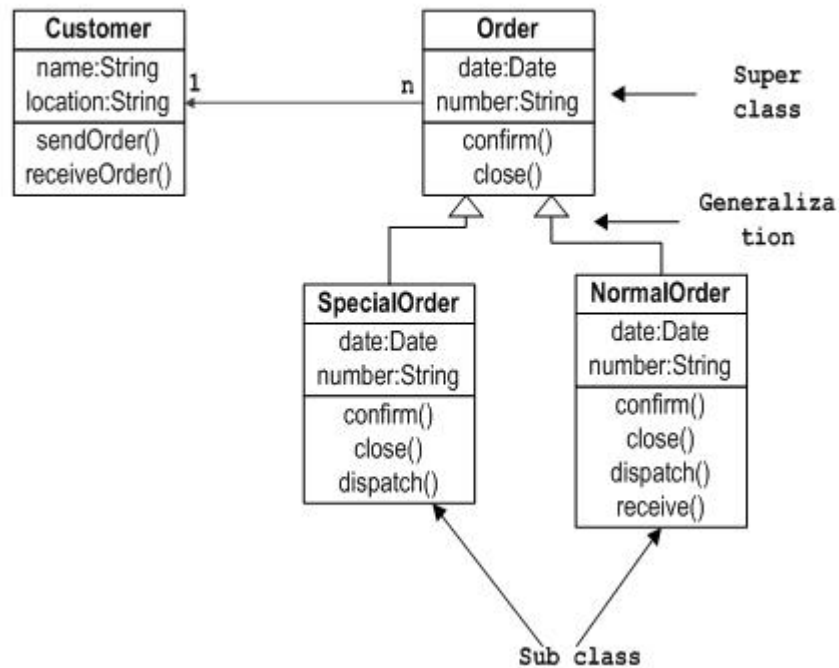
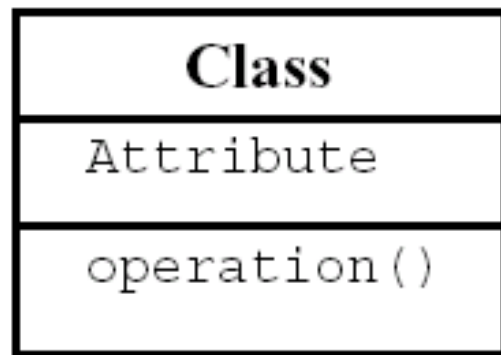


Class diagrams

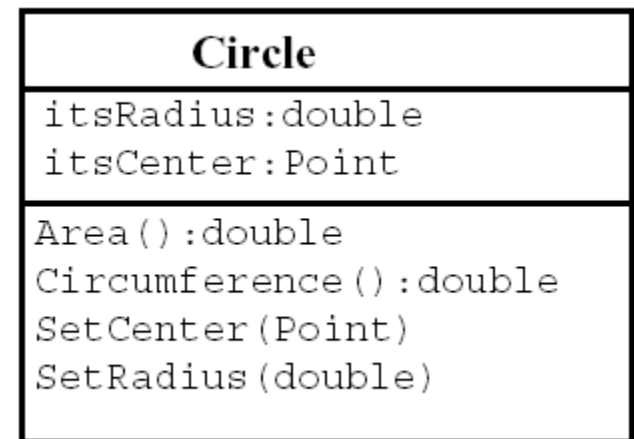


The Class Diagram Notation

- Identify classes, attributes of each class, and operations of each class
- Classes, their attributes and methods are specified based on the objects needed to realized use case and interfaces to external entities

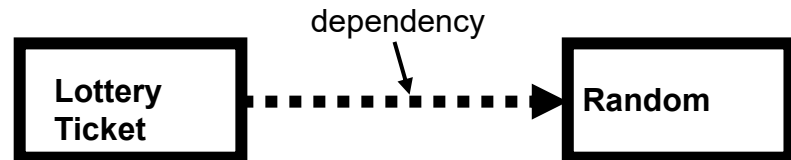
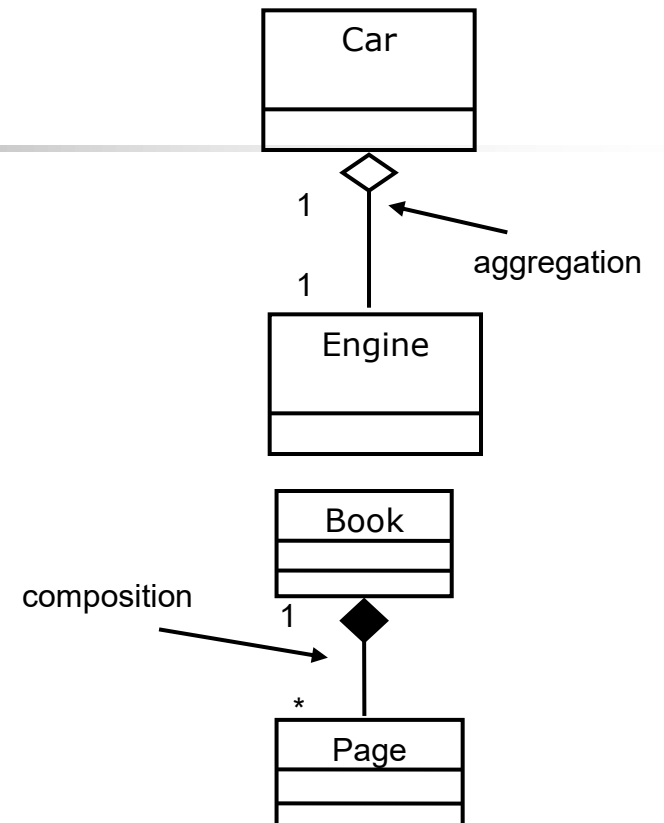


Detailed
Attributes,
Data types,
And operations
Are defined/
refined
During design

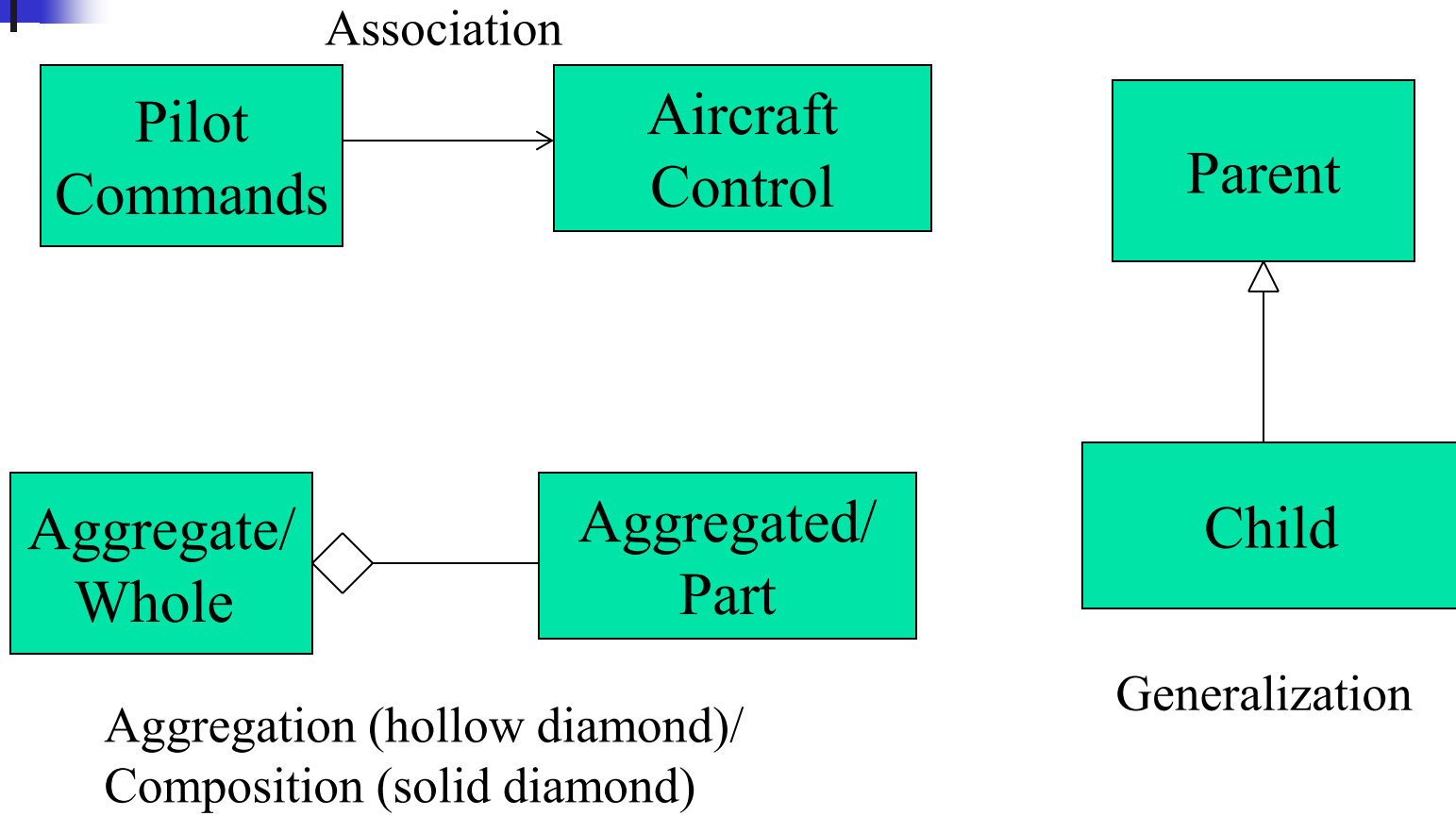


Association types

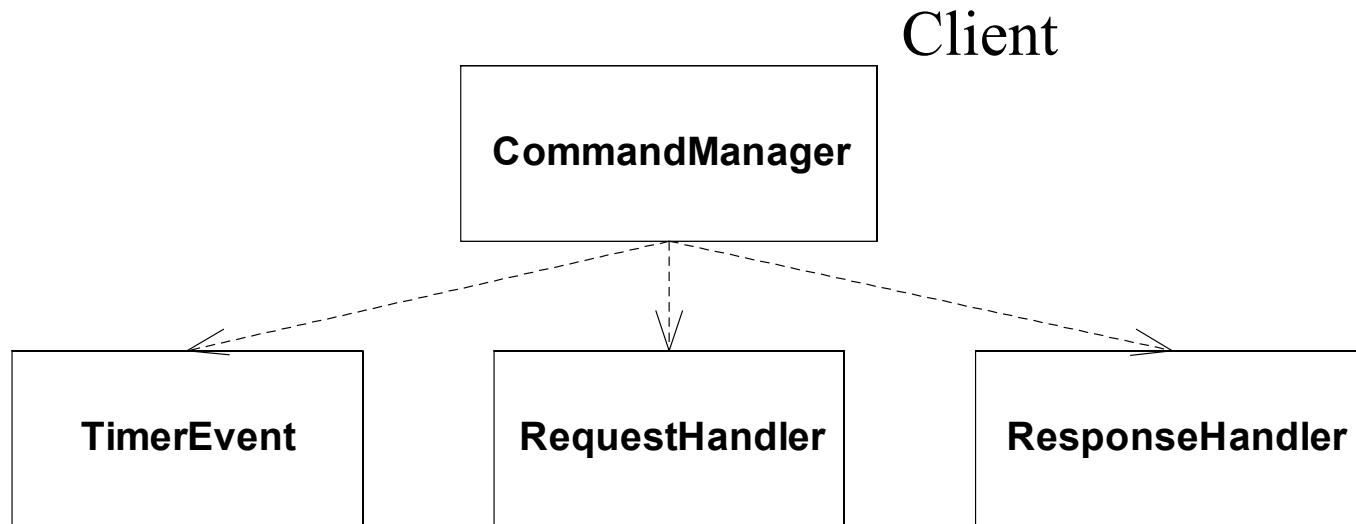
- **aggregation:** "is part of"
 - symbolized by a clear white diamond
- **composition:** "is entirely made of"
 - stronger version of aggregation
 - the parts live and die with the whole
 - symbolized by a black diamond
- **dependency:** "uses temporarily"
 - symbolized by dotted line
 - often is an implementation detail, not an intrinsic part of that object's state



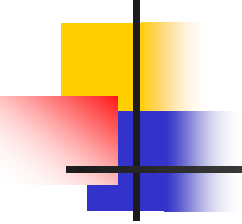
Identify Class relationships



Dependency: A Special Case of Association



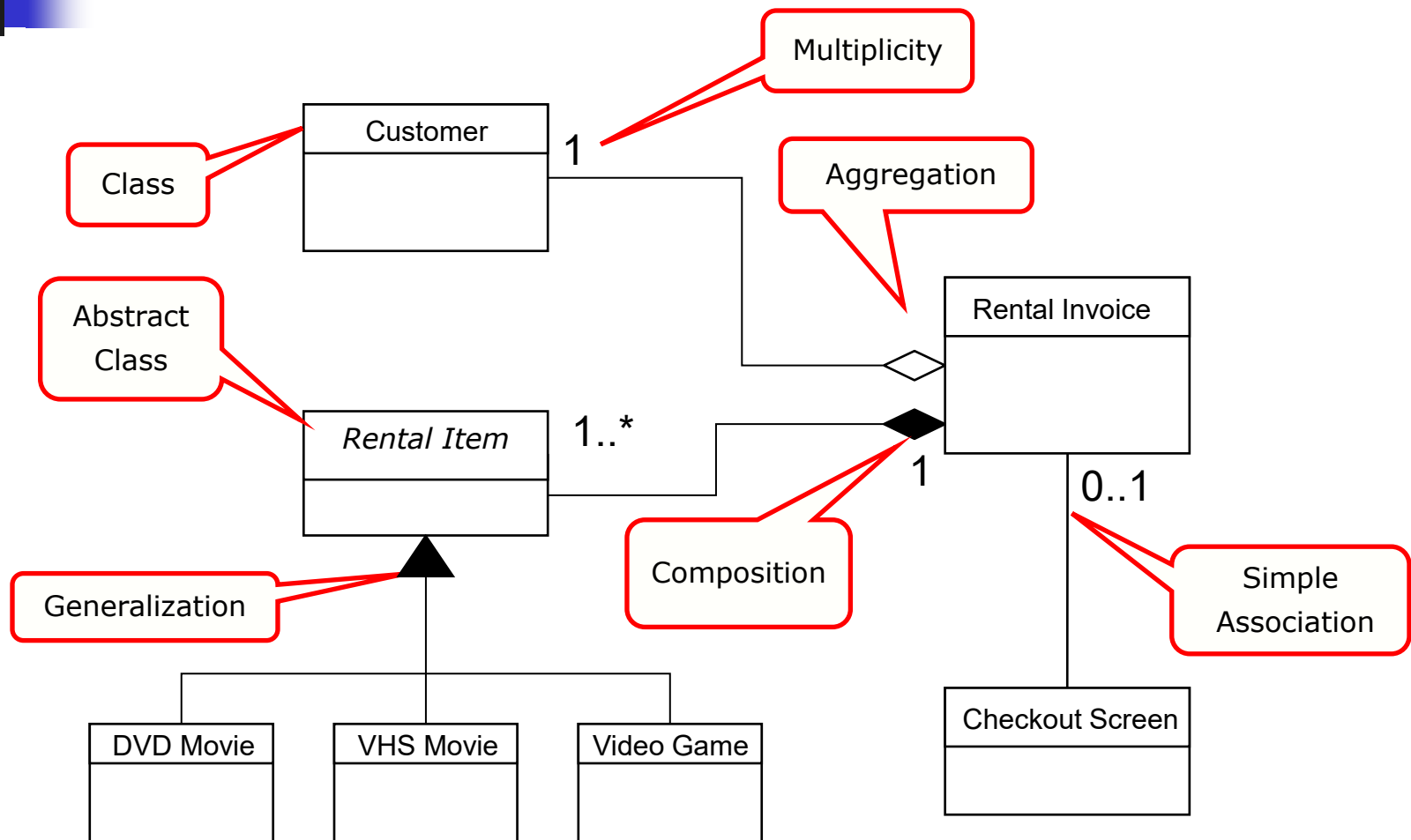
CommandManager (Client class) depends on services provided by the other three server classes



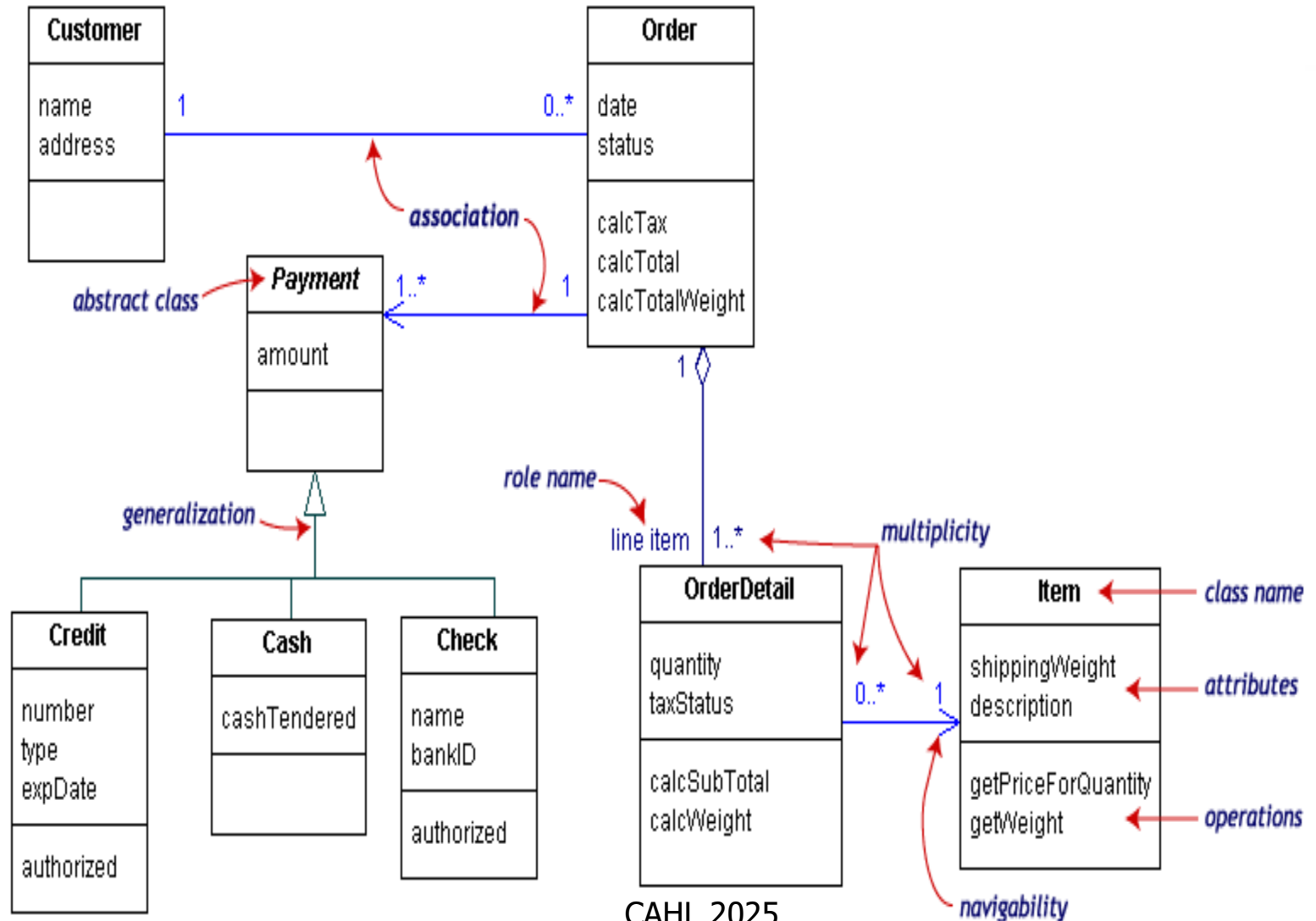
Multiplicity of Relationships

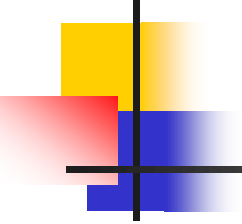
Multiplicities	Meaning
0..1	zero or one instance. The notation <i>n</i> . . <i>m</i> indicates <i>n</i> to <i>m</i> instances.
0..* or *	no limit on the number of instances (including none).
1	exactly one instance
1..*	at least one instance

Eksempel



Example of identifying Class Relations, Multiplicities, Attributes and operations





Visibility på klassediagram

Symbol	Access Modifier
+	public
-	private
#	protected
~	package
/	derived

Eksempel på klassediagram

Visual Paradigm Standard(cahl(KEA))

